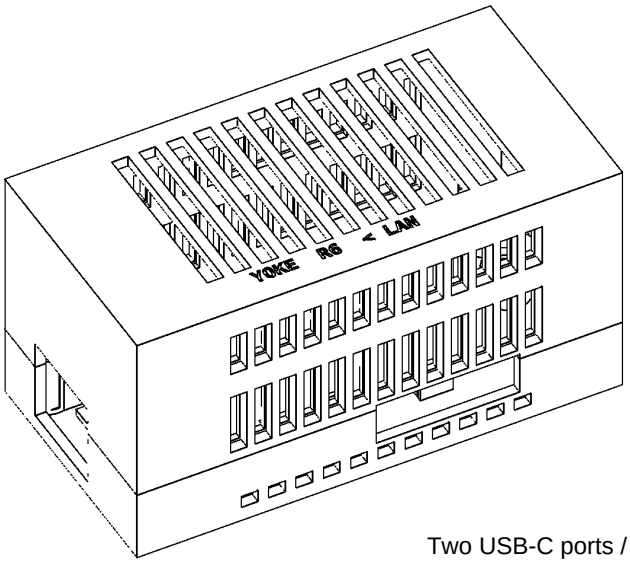
Know the parts and hardware

For the ordered ZERO 3E 2GB WITH GPIO (Amazon B0CZD1C54N) and Radxa PoE HAT (B0D1KF1L7G). Electronics have not arrived. Print and unloaded checks can proceed; verify the delivered variant before fitting or powering.

89 x 45.5 x 46 | two shells + two removable PCB clips

LAN end



Two USB-C ports / one low side opening

Opposite end closed. HDMI covered. microSD: open case and lift SBC.

1 Inventory and measure owner stock

Count four printed parts: base, lid, front clip and right clip. Check four M3 x 16 case screws, two M3 x 8 clip screws and six RX-M3x5.7 inserts. Use a caliper/depth gauge to check actual lengths, head envelopes and insert stock identity before heating.

Stop for wrong stock, oversized heads, damaged threads or inaccessible tools.

Nominal screws: M3 x 16 (4) and M3 x 8 (2), measured under head; each head at most diameter 6 x height 3. Inserts: RX-M3x5.7 (6), diameter 4.6 x length 5.7. Printed pilots: diameter 4.0. Do not substitute by thread label alone.

Separate factory hardware: 4 Radxa insulated copper pillars + 8 supplied CM2.5x4 screws. These conductive joints remain electrically functional; plastic retention touches PCB edges only.

Tools: caliper/depth gauge, matching socket driver, correct Radxa screw driver, controlled insert heater/tip, nonflammable support, actual cables, temperature probe and CPU monitoring. Confirm tool access on empty parts.

Source allowance: larger official RS109_RA023 assembly (18 mm spacing) gives 3.5 mm nominal roof clearance. Earlier 14.5/18 mm spacing discrepancy remains unresolved. Diameter-8 corner clearances and headroom are design allowances, not measurements of delivered stock.

Fit empty; heat-set six inserts

2 Test the empty shell pair first

With no electronics, inserts or screws, match the LAN ends and lower the lid vertically. It must seat evenly by hand, then lift off freely. Clear loose print debris only. Check the actual driver through all six screw wells and the heater at all six pilots.

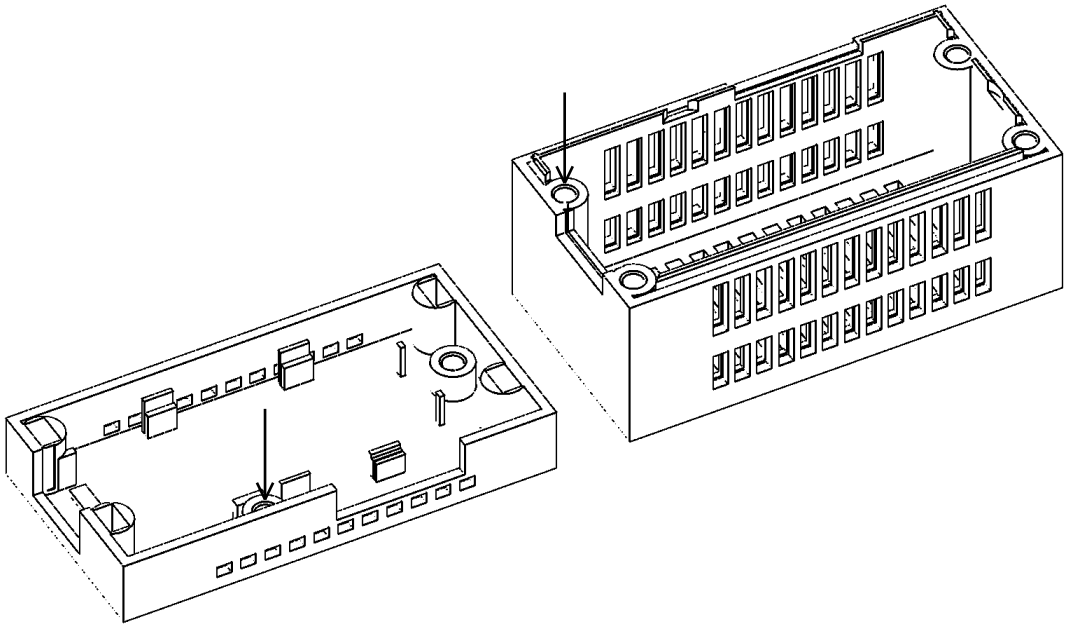
Stop for binding or skew. Screws must never pull a poor fit closed.

3 Heat only the empty plastic parts

Support the EMPTY lid roof-down and EMPTY base floor-down on a flat nonflammable surface. Heat-set four inserts into the lid boss mouths and two into the base clip bosses, straight down and flush with the mouth. Follow the insert maker's controlled-heater method. Cool fully; inspect flushness, threads and distortion.

Stop for a tilted or proud insert, crack, lifted boss or blocked thread. Never heat plastic with electronics installed.

BASE: floor down **LID: roof down**



2 inserts / clip boss mouths **4 inserts / shell boss mouths**

The illustrated lid is inverted from its assembled orientation. Its insert mouth is the seam face. Heat-set the insert end flush with that face; in assembly it occupies Z = 13.5 to 19.2. Base inserts finish flush at the clip boss top, Z = 8.0, and extend to Z = 2.3.

Let both parts cool completely before threading screws or installing electronics. Inspect all six bosses and threads. Test actual driver and heater clearance; the model's tool envelopes are not measurements of your tools.

Empty nested joint: 0.5 clearance per side, 2.0 engagement, 1.2 skirt thickness and 0.5 entry chamfer. Both mating edges print above the bed. Lower and lift vertically; there is no hook or sideways latch motion.

After cooling, repeat the empty fit. Check the two loose clips seat flat on their own bosses. Never drive a screw to reshape a boss, flatten a tilted insert or close an uneven seam.

SBC first, then clips, then HAT

4 Fit factory pillars outside the case

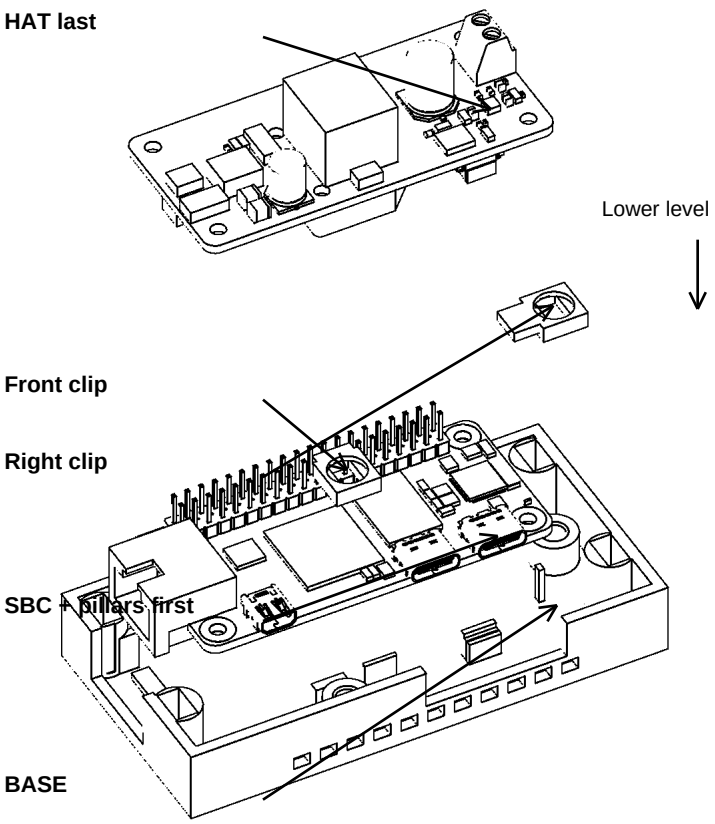
Disconnect all power and cables. Outside the case, fit the four Radxa lower screws and four supplied insulated copper pillars to the SBC, following Radxa's assembly instructions. Fit microSD now. Preserve the factory electrical contact faces; do not replace these screws with case screws or add insulation at their contacts.

Stop if the delivered SBC/HAT variant, pillars or electrical joints differ from the reference.

5 Lower SBC with HAT absent

With HAT ABSENT, lower the SBC and attached pillars level into the base. Ethernet faces the LAN end; both USB-C sockets face the single low side opening. The PCB rests on four edge pads; all components and factory hardware must clear.

Stop for rocking, bowed PCB, caught GPIO pins, or a component bearing on the case.



6 Screw the two clips down from above

Still with HAT ABSENT, lower the front clip onto its boss by the USB-C side and the right clip onto the opposite-end boss. Their toes face inward over the PCB edges. Start both M3 x 8 screws from above and lightly snug. Each clip seats on its plastic boss, leaving nominal 0.4 mm vertical PCB play.

Stop for PCB clamping, an unseated clip, early screw bottoming or a turning insert.

7 Lower HAT and fit four upper Radxa screws

Align the HAT GPIO connector and lower it level onto the SBC and supplied pillars. Tighten the four upper Radxa screws as its instructions require. Confirm full connector seating and intact conductive joints. The clips are already installed; do not try to insert them under an assembled HAT.

Stop if HAT, fan, cable or factory screw contacts the enclosure or clip hardware.

Lid omitted here; see page 4. Official assembly meshes are raised apart for sequence clarity. Supplied pillars and screws must be installed as stated even where not separately represented in the source CAD. Fan shown is a reference feature; its inclusion in the ordered HAT is unconfirmed.

Factory assembly: docs.radxa.com/en/zero/zero3/accessories/3e-poe-hat

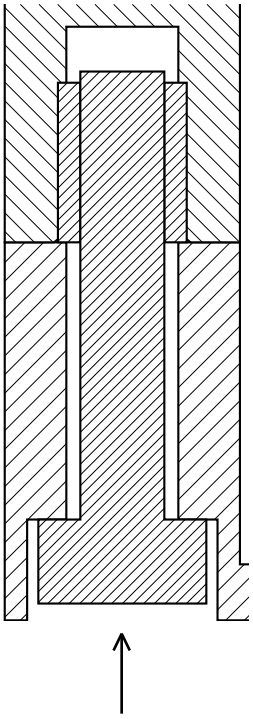
Close freely; screw from below

8 Close, hold together, invert, fasten

Match LAN ends and lower the lid straight down over the completed module. It must seat freely with an even seam. Hold the two closed halves securely, invert together, and start all four M3 x 16 screws from the base underside. Lightly snug opposite corners in turn, then return floor-down.

Stop for roof contact, binding, early bottoming, cross-threading, spinning inserts or proud screw heads. Do not use screw force to close the seam.

A-A / case screw / Y = 1.6



21.2 blind pilot end

19.6 screw tip

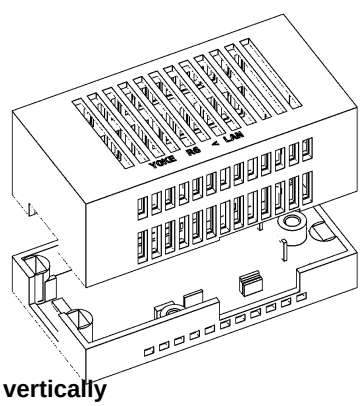
19.2 insert top

13.5 seam / insert mouth

3.6 head bearing land

0.6 head bottom / recess

Driver from base underside



Lid lowers vertically

Four M3 x 16 joints: 5.7 nominal full insert overlap; screw tip extends 0.4 beyond the insert into 2.0 of blind-pilot reserve (1.6 remains). Inspect actual lengths: a long screw must not bottom out before the seam seats.

Load path: screw head -> base bearing land -> base/lid plastic stop at seam -> lid insert threads -> screw. Neither PCB is in this clamp stack.

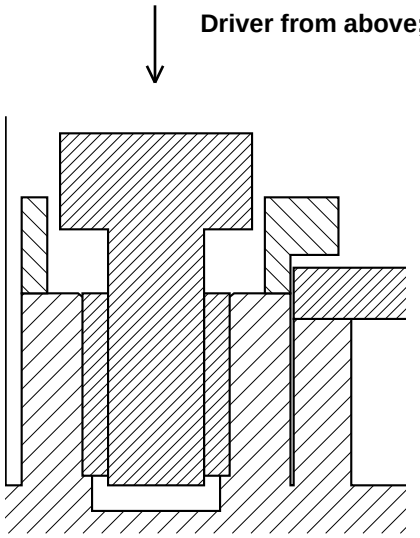
Section is an actual cut through the released base, lid, screw and insert STL meshes, not a schematic block. Base and lid have opposing hatches; metal has close hatching. Threads are not modeled.

The arrows show access in the assembled coordinate frame. For fastening, hold both shells together and turn the case upside down so the driver is above the bottom wells. Start all four screws before snugging opposite corners.

Hand-tight only: no enclosure torque has been validated. Heads should lie 0.6 below the bottom surface with nominal stock. Stop for a proud head, spinning insert, bowed case or any change in PCB support.

Retention and reverse service

B-B / front clip screw / X = 23.5 / Y horizontal, Z up

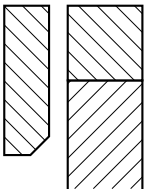


M3 x 8 screw: head bears at Z = 10; head top 13; tip 2.0. Insert spans Z = 2.3 to 8.0. Full 5.7 insert overlap; 0.8 remains to blind pilot bottom Z = 1.2.

Clip underside seats on its plastic boss at Z = 8. The toe underside is Z = 9.2, above PCB top Z = 8.8. Nominal play 0.4; lower PCB rests on Z = 7.2 pads.

Load path: screw head -> clip land -> base boss -> insert threads. Toe limits PCB lift; it must not clamp the board or share the factory pillar load.

C-C / nested rear wall / X = 22



Actual STL joint cut: lid skirt inside the base wall, 0.5 side gap, 2.0 engagement. Matching parts lift apart vertically without screw force. Vents also pass through the inner skirt.

9 Power off; reverse the installation order

Disconnect power and all cables. Hold the closed case inverted, remove all four bottom M3 x 16 screws, return floor-down and lift the lid vertically. For microSD or SBC removal: undo the four upper Radxa screws, lift the HAT level off GPIO, remove both M3 x 8 clip screws and clips, then lift the SBC with its pillars. Reassemble in page 3 order.

Do not pry on a PCB or connector. Do not pull the SBC with clips or HAT still fitted.

For lid-only inspection, the HAT and clips can remain fitted. For microSD, lift the SBC after HAT and clip removal; no outside microSD opening is provided. Keep removed factory screws separate from M3 case hardware. Do not loosen the lower conductive joints merely to open the case.

Record first-article checks

All physical checks below: NOT RUN. Complete after parts arrive.

A Inspect unpowered fit and cable access

Unpowered: record stock measurements; inspect the free empty seam, cooled inserts, PCB support and toe clearance. Fit actual Ethernet and both USB-C plugs and collars. Gently turn the closed case through ordinary use orientations and record play/rattle. Check cable strain and every factory fastener remains clear.

Stop for forced plugs, cable strain, loose inserts, PCB bowing, component contact or changed support.

B Prove the service sequence ten times

Unpowered and unplugged, perform ten complete service/reassembly cycles, including HAT and clip removal. Each lid must seat freely before screws. Record cycle number, seam, thread condition, insert movement and PCB play; check actual microSD access with the SBC lifted.

Stop at the first new bind, damaged thread, loose insert or cracked plastic; return that interface to design.

C Compare open and closed temperatures

Only after unpowered checks pass, use the intended PoE supply and workload on a hard flat desk, vents clear. Record ambient and CPU temperature, case surface at several points, elapsed time, throttling and resets every 5 minutes. Compare lid-open and lid-closed runs at the same ambient, workload and fan state, at least 30 minutes each and until three successive readings change by less than 1 C. Record fan presence and operation; allow cooling between runs.

Stop for throttling, unexpected resets, odor, plastic softening, any component maker's limit, or case surface above 45 C. The 45 C threshold is a provisional PLA first-article limit, not a manufacturer-qualified rating. Lack of a temperature plateau leaves the test incomplete.

Air paths: low side inlets, inter-board and upper side rows, and roof slots. Keep them clear on a hard desk. A fan is optional and its presence must be recorded. Vent area and 3.5 roof clearance do not establish cooling performance or a safe continuous-power rating.

Record before deciding to use this enclosure

Date / printed material / measured stock: _____

Delivered board + HAT / fan / fit + cable result: _____

Cycles / play / inserts / failures: _____

Ambient / workload / open vs closed temperatures / time: _____

Outcome / incomplete checks / next design change: _____